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(54) **ELECTRODE, BATTERY, AND POWER
CONSUMING DEVICE**

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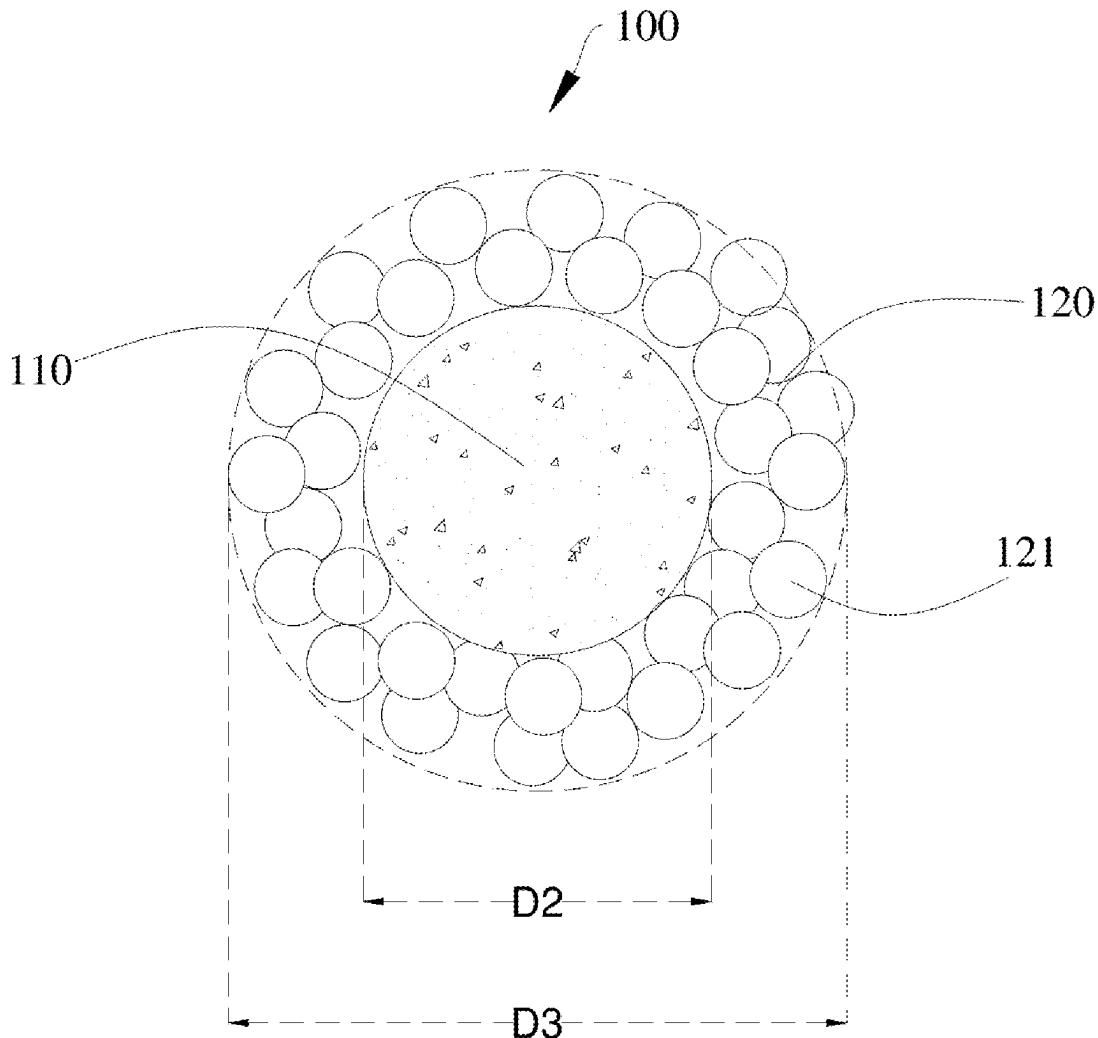
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(57)

ABSTRACT

An electrode, a battery, and a power consuming device are provided. The electrode includes a current collector layer and an active layer. The active layer is disposed on a surface of the current collector layer. The active layer includes active particles and electrolyte retention particles. Each of the electrolyte retention particles includes a core and a shell. The core is coated with the shell. The shell includes multiple sub-particles. Each of the multiple sub-particles has capillary pores.



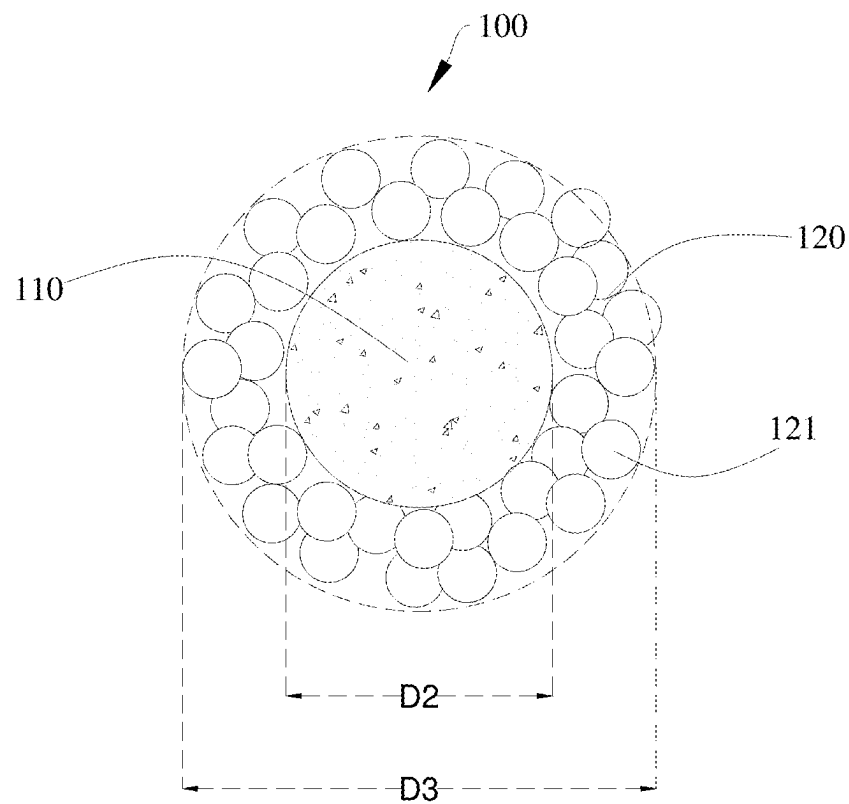


FIG. 1

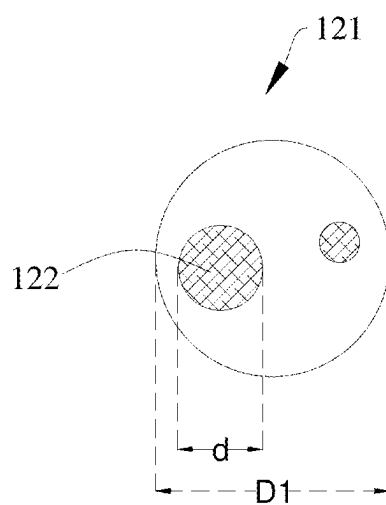


FIG. 2

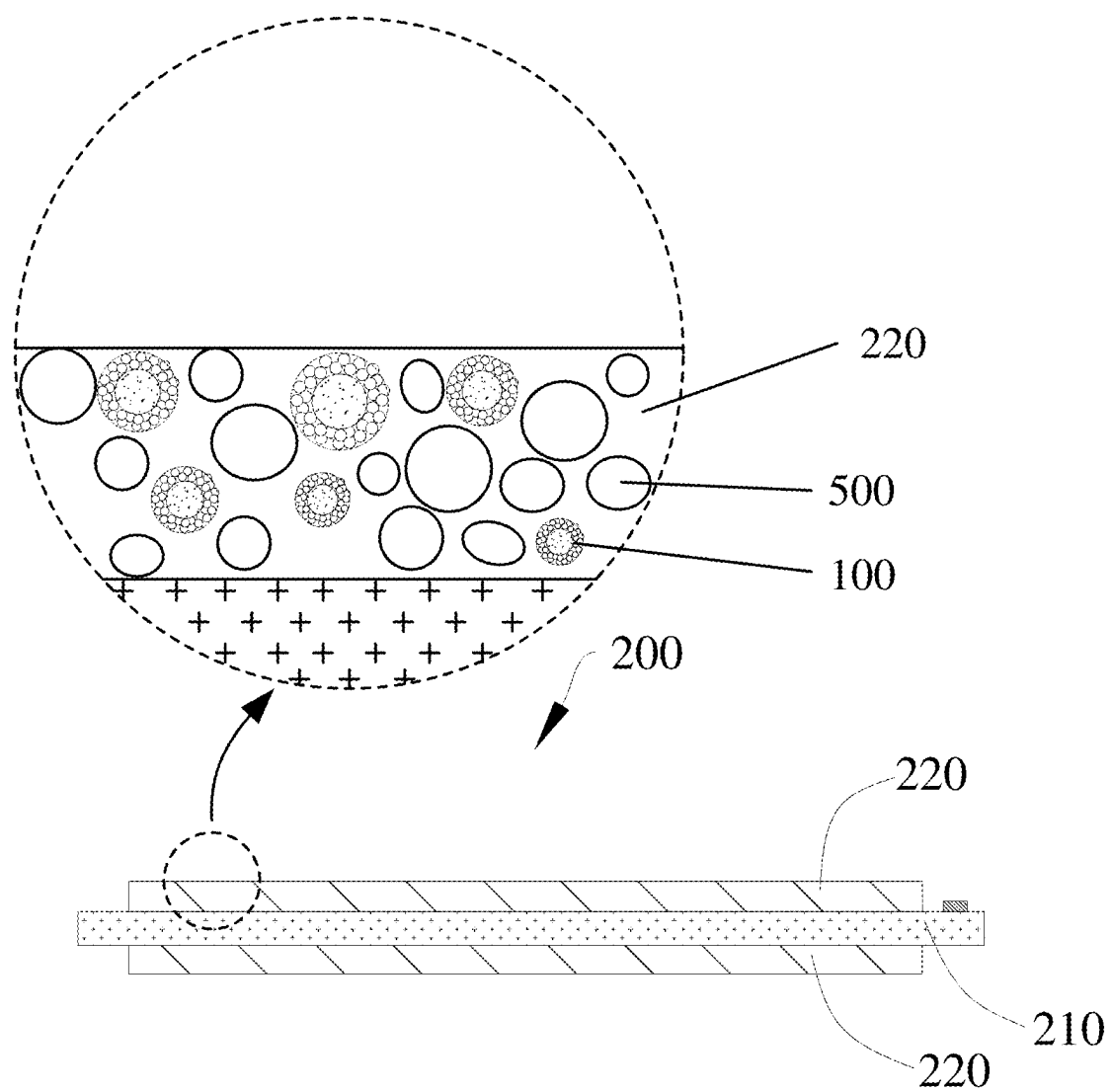


FIG. 3

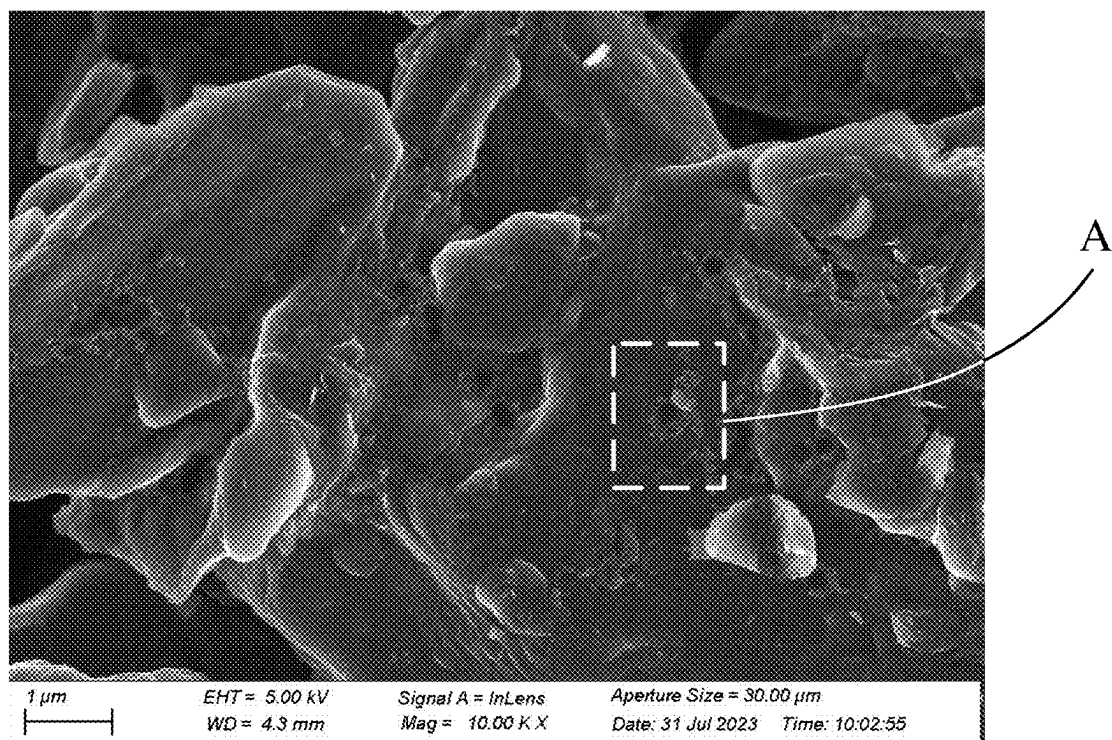


FIG. 4

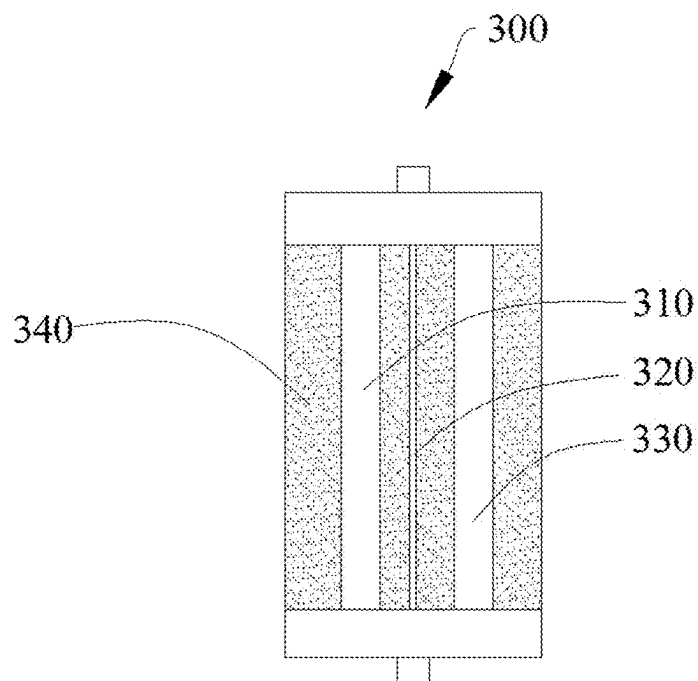


FIG. 5

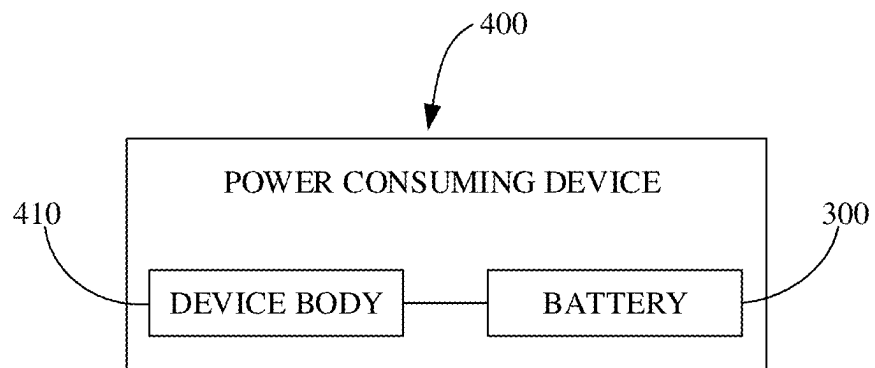


FIG. 6

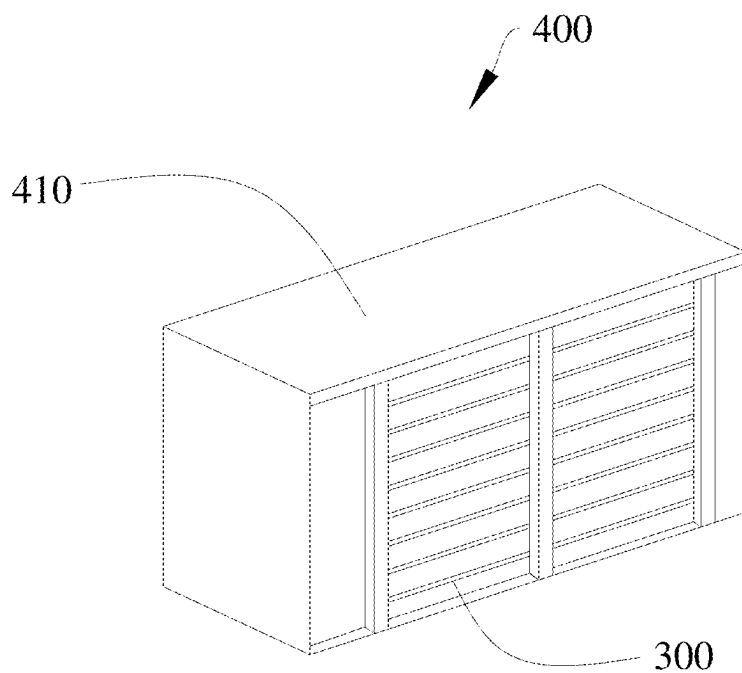


FIG. 7

ELECTRODE, BATTERY, AND POWER CONSUMING DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority under 35 U.S.C. § 119 (a) to Chinese Patent Application No. 202410250903.9, filed Mar. 5, 2024, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure relates to the field of battery technology, and in particular to an electrode, a battery, and a power consuming device.

BACKGROUND

[0003] During charging and discharging of batteries, active ions in the electrolyte infiltrate into electrode materials of electrodes and then react with active substances therein, thereby releasing electrons. If the infiltrability of the electrode for the electrolyte is poor, it may result in electrolyte loss or uneven distribution during charging and discharging, subsequently affecting the energy efficiency and cycle capacity retention rate of the battery. Therefore, improvement of the electrolyte retention capability of the electrode is one of the key approaches to improvement of the battery performance.

SUMMARY

[0004] In a first aspect, an electrode is provided in the present disclosure. The electrode includes a current collector layer and an active layer. The active layer is disposed on a surface of the current collector layer. The active layer includes active particles and electrolyte retention particles. Each of the electrolyte retention particles includes a core and a shell. The core is coated with the shell. The shell includes multiple sub-particles. Each of the multiple sub-particles has capillary pores.

[0005] In a second aspect, a battery is provided in the present disclosure. The battery includes a positive electrode, a separator, a negative electrode, and an electrolyte. The separator is disposed at one side of the positive electrode. The negative electrode is disposed at one side of the separator positioned facing away from the positive electrode. At least one of the positive electrode or the negative electrode is an electrode. The electrode includes a current collector layer and an active layer. The active layer is disposed on a surface of the current collector layer. The active layer includes active particles and electrolyte retention particles. Each of the electrolyte retention particles includes a core and a shell. The core is coated with the shell. The shell includes multiple sub-particles. Each of the multiple sub-particles has capillary pores. At least a part of the positive electrode and at least a part of the negative electrode are immersed in the electrolyte.

[0006] In a third aspect, a power consuming device is provided in the present disclosure. The power consuming device includes a device body and a battery. The battery is configured to supply power to the device body. The battery includes a positive electrode, a separator, a negative electrode, and an electrolyte. The separator is disposed at one side of the positive electrode. The negative electrode is disposed at one side of the separator positioned facing away

from the positive electrode. At least one of the positive electrode or the negative electrode is an electrode. The electrode includes a current collector layer and an active layer. The active layer is disposed on a surface of the current collector layer. The active layer includes active particles and electrolyte retention particles. Each of the electrolyte retention particles includes a core and a shell. The core is coated with the shell. The shell includes multiple sub-particles. Each of the multiple sub-particles has capillary pores. At least a part of the positive electrode and at least a part of the negative electrode are immersed in the electrolyte.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] In order to explain technical solutions in embodiments of the present disclosure more clearly, the following will give a brief introduction to the accompanying drawings required for describing the implementations. Apparently, the accompanying drawings in the following description illustrate some implementations of the present disclosure. For those of ordinary skill in the art, other accompanying drawings can be obtained according to these accompanying drawings without creative efforts.

[0008] FIG. 1 is a schematic structural view of an electrolyte retention particle according to an embodiment of the present disclosure.

[0009] FIG. 2 is a schematic structural view of a sub-particle according to an embodiment of the present disclosure.

[0010] FIG. 3 is a schematic structural view of an electrode according to an embodiment of the present disclosure.

[0011] FIG. 4 is an electron microscope image of an electrode according to an embodiment of the present disclosure.

[0012] FIG. 5 is a schematic structural view of a battery according to an embodiment of the present disclosure.

[0013] FIG. 6 is a block view of a circuit of a power consuming device according to an embodiment of the present disclosure.

[0014] FIG. 7 is a schematic structural view of a power consuming device according to an embodiment of the present disclosure.

[0015] Description of reference signs of the accompanying drawings: 100—electrolyte retention particle, 110—core, 120—shell, 121—sub-particle, 122—capillary pore, 200—electrode, 210—current collector layer, 220—active layer, 300—battery, 310—positive electrode, 320—separator, 330—negative electrode, 340—electrolyte, 400—power consuming device, 410—device body, 500—active particle.

DETAILED DESCRIPTION

[0016] The following will clearly and completely describe technical solutions of embodiments of the present disclosure with reference to the accompanying drawings in embodiments of the present disclosure. Apparently, embodiments described herein are merely some embodiments, rather than all embodiments, of the present disclosure. Based on the embodiments of the present disclosure, all other embodiments obtained by those of ordinary skill in the art without creative effort shall fall within the protection scope of the present disclosure.

[0017] The terms such as “first”, “second”, etc., in the specification, the claims, and the above accompanying drawings of the present disclosure are used to distinguish differ-

ent objects, rather than describing a particular order. In addition, the terms “including”, “comprising”, and “having” as well as variations thereof are intended to cover non-exclusive inclusion. For example, a process, method, system, product, or device including a series of steps or units is not limited to the listed steps or units, on the contrary, it may alternatively include other steps or units that are not listed; alternatively, other steps or units inherent to the process, method, product, or device may be included either.

[0018] The term “embodiment” or “implementation” referred to herein means that particular features, structures, or properties described in conjunction with embodiments or implementations may be defined in at least one embodiment of the present disclosure. The phrase “embodiment” or “implementation” appearing in various places in the specification does not necessarily refer to the same embodiment or an independent/alternative embodiment that is mutually exclusive with other embodiments. Those of ordinary skill in the art will understand expressly and implicitly that an embodiment described in the present disclosure may be combined with other embodiments.

[0019] During charging and discharging of a battery, active ions in an electrolyte infiltrate into an electrode material of an electrode, and react with active substances therein, thereby releasing electrons. If an infiltrability of the electrode for the electrolyte is poor, it may result in electrolyte lost or unevenly distribution during charging and discharging, subsequently affecting the energy efficiency and cycle capacity retention rate of the battery. Therefore, improvement of the electrolyte retention capability of the electrode is one of the key approaches to improvement of the battery performance.

[0020] Commonly, the porosity of the electrode is increased by means such as increasing the pressure of the roller to improve the electrolyte retention capability of the electrode. However, it is difficult to uniformly control the pore size, leading to poor stability of the electrodes between batches. In addition, high-swelling binders are added to the electrode to define more channels, thereby improving the electrolyte retention capability. However, excessive swelling may increase internal stress in the battery, which is detrimental to the subsequent cycle of the battery.

[0021] As illustrated in FIG. 1 to FIG. 4, an electrode 200 is provided in the present disclosure. The electrode 200 includes a current collector layer 210 and an active layer 220. The active layer 220 is disposed on a surface of the current collector layer 210. The active layer 220 includes active particles 500 and electrolyte retention particles 100. Each of the electrolyte retention particles 100 includes a core 110 and a shell 120. The core 110 is coated with the shell 120. The shell 120 includes multiple sub-particles 121. Each of the multiple sub-particles 121 has capillary pores 122.

[0022] It may be understood that, the active layer 220 and the current collector layer 210 are stacked.

[0023] Alternatively, the electrode 200 may be a positive electrode 310 or a negative electrode 330.

[0024] Alternatively, in some embodiments, the active layer 220 is disposed on one surface of the current collector layer 210. In other embodiments, two active layers 220 are respectively disposed on two opposite surfaces of the current collector layer 210.

[0025] It may be understood that, the electrolyte retention particles 100 is applied to the electrode 200 assembled into

the battery 300. The battery 300 includes the electrode 200 and an electrolyte 340. At least a part of the electrode 200 is immersed in the electrolyte 340.

[0026] It may be understood that, the electrolyte retention particle 100 has a core-shell structure.

[0027] In this embodiment, the electrode 200 includes the current collector layer 210 and the active layer 220 that are stacked, and the active layer 220 is disposed on the surface of the current collector layer 210. Therefore, the current collector layer 210 can carry the active layer 220 and collect and output currents generated by the active particles 500, thereby realizing the concentration and transmission of the currents. In this embodiment, the active layer 220 includes the active particles 500 and the electrolyte retention particles 100. Each electrolyte retention particle 100 includes the core 110 and the shell 120 coating the core 110. The shell 120 is composed of the multiple sub-particles 121 that are jointed together. Each sub-particle 121 has the capillary pores 122. When the electrode 200 is assembled into the battery 300, each capillary pore 122 can absorb electrolyte 340, so that the shell 120 of the electrolyte retention particle has multiple “antennae”, that is, the capillary pore 122 of the shell 120 acts like a straw to absorb the electrolyte 340 and store the electrolyte on the core 110. Therefore, the electrolyte retention particles 100 have a high electrolyte retention performance, and the storage capacity of the electrode for the electrolyte is improved, so that the electrode 200 has a good infiltrability for the electrolyte. When the battery 300 is charged or discharged, the active particles 500 in the electrode 200 can be in effective contact with the electrolyte 340. The active ions in the electrolyte 340 can react chemically with the active particles 500 in the active layer 220 and then release electrons. Therefore, the electrode 200 has the good electrolyte retention performance, thereby improving the cycle performance and energy efficiency of the battery 300 when the electrode 200 is applied to the battery 300.

[0028] It may be understood that, in terminology of the present disclosure, “infiltrability” refers to an interaction capability between the active material in the electrode 200 and the electrolyte 340.

[0029] It may be understood that, FIG. 4 is an electron microscope image of an electrode 200 according to an embodiment of the present disclosure. A region indicated by a dotted-line box in FIG. 4 shows an electrolyte retention particle 100 disposed on the electrode 200. The electrolyte retention particle 100 includes a core 110 and a shell 120. Due to different materials, the shell 120 of the electrolyte retention particle 100 and the core 110 of the electrolyte retention particle 100 exhibit color differences in the microscope image.

[0030] In some embodiments, each of the capillary pores 122 has a radial dimension d satisfying: $2\text{ nm} \leq d \leq 30\text{ nm}$.

[0031] Specifically, the radial dimension d of the capillary pore 122 may be, but is not limited to, 2 nm, 2.5 nm, 3 nm, 5 nm, 6 nm, 8 nm, 9 nm, 10 nm, 12 nm, 15 nm, 16 nm, 18 nm, 19 nm, 20 nm, 22 nm, 23 nm, 24 nm, 25 nm, 26 nm, 27 nm, 28 nm, 29 nm, 30 nm, etc.

[0032] In this embodiment, when the radial dimension d of the capillary pore 122 satisfies: $2\text{ nm} \leq d \leq 30\text{ nm}$, the radial dimension of the capillary pore 122 is in a reasonable range, and the adsorption capacity and storage capacity of the capillary pore 122 for the electrolyte 340 are both in a reasonable range, so that a liquid adsorption amount per unit particle is relatively high. When the electrode 200 is

assembled into the battery 300, the electrode 200 has the good infiltrability for the electrolyte 340, which helps to reduce the interface impedance between the electrode 200 and the electrolyte 340, and further helps to improve the cycle performance and energy efficiency of the battery 300. In addition, excessively large radial dimension of the capillary pore 122 can be avoided to prevent a reduction in the structural strength of the sub-particle 121, so that the electrode 200 has good structural strength. When the radial dimension of the capillary pore 122 is too large, the capillary pore 122 occupies an excessive portion of the space in the sub-particle 121, resulting in reduced structural strength of the sub-particle 121. Since the shell 120 is composed of the multiple sub-particles 121 that are jointed together, the connections between the sub-particles 121 are also weakened accordingly, further reducing the structural strength of the shell 120. When the electrode 200 is assembled into the battery 300, the electrolyte retention particle 100 with a weaker structural strength is prone to deformation or fracturing during charging or discharging of the battery 300, and then the overall structural strength of the electrode 200 is reduced. Moreover, the electrolyte 340 is unable to be effectively fixed and maintained, so that the infiltrability of the electrode 200 for the electrolyte 340 is reduced, thereby reducing the cycle performance and energy efficiency of the battery 300. When the radial dimension of the capillary pore 122 is too small, the adsorption capacity and storage capacity of the capillary pore 122 for the electrolyte 340 are reduced, so that the liquid adsorption amount per unit particle is reduced. When the electrode 200 is assembled into the battery 300, the infiltrability of the electrode 200 for the electrolyte 340 is reduced, and the interface impedance at which the active ions in the electrolyte 340 react with the active particles 500 in the electrode 200 is increased, thereby reducing the cycle performance and energy efficiency of the battery 300.

[0033] Preferably, in some embodiments, the radial dimension d of the capillary pores 122 satisfies: $5\text{ nm} \leq d \leq 25\text{ nm}$.

[0034] Specifically, the radial dimension d of the capillary pore 122 may be, but is not limited to, 5 nm, 6 nm, 8 nm, 9 nm, 10 nm, 12 nm, 15 nm, 16 nm, 18 nm, 19 nm, 20 nm, 22 nm, 23 nm, 24 nm, 25 nm, etc.

[0035] In some embodiments, in the shell 120, the capillary pores 122 have a distribution density a satisfying: $0.4\text{ cm}^3/\text{g} \leq a \leq 3\text{ cm}^3/\text{g}$.

[0036] Specifically, the distribution density a of the capillary pores 122 may be, but is not limited to, $0.4\text{ cm}^3/\text{g}$, $0.6\text{ cm}^3/\text{g}$, $0.8\text{ cm}^3/\text{g}$, $0.9\text{ cm}^3/\text{g}$, $1\text{ cm}^3/\text{g}$, $1.2\text{ cm}^3/\text{g}$, $1.5\text{ cm}^3/\text{g}$, $1.6\text{ cm}^3/\text{g}$, $1.8\text{ cm}^3/\text{g}$, $1.9\text{ cm}^3/\text{g}$, $2\text{ cm}^3/\text{g}$, $2.2\text{ cm}^3/\text{g}$, $2.5\text{ cm}^3/\text{g}$, $2.6\text{ cm}^3/\text{g}$, $2.7\text{ cm}^3/\text{g}$, $2.9\text{ cm}^3/\text{g}$, $3\text{ cm}^3/\text{g}$, etc.

[0037] In this embodiment, when the distribution density a of the capillary pores 122 satisfies: $0.4\text{ cm}^3/\text{g} \leq a \leq 3\text{ cm}^3/\text{g}$, the distribution density of the capillary pores 122 is in a reasonable range, so that the adsorption capability and the infiltrability of the electrolyte retention particle 100 for the electrolyte 340 are both in a reasonable range. When the electrode 200 is assembled into the battery 300, the electrode 200 has the good infiltrability for the electrolyte 340, and an impedance at which the active ions in the electrolyte 340 chemically react with the active particles 500 in the electrode 200 is effectively reduced, so that the battery 300 has the higher cycle performance and energy efficiency. In addition, excessive volume of the shell 120 occupied due to

excessive distribution density of the capillary pores 122 can be avoided, so that the sub-particle 121 and the electrolyte retention particle 100 have the good structural strength, and thus the overall structural strength of the electrode 200 is good. When the distribution density of the capillary pores 122 is too large, the volume occupied by the capillary pores 122 in the shell 120 is too large, so that the structural strength of the shell 120 is weak. When the electrode 200 is assembled into the battery 300, the electrolyte retention particle 100 with a weaker structural strength is prone to deformation or fracturing during charging or discharging of the battery 300, and then the overall structural strength of the electrode 200 is reduced. Moreover, the electrolyte 340 is unable to be effectively fixed and maintained, so that the infiltrability of the electrode 200 for the electrolyte 340 is reduced, thereby reducing the cycle performance and energy efficiency of the battery 300. When the distribution density of the capillary pores 122 is too small, in other words, when in the shell 120, the volume occupied by all the capillary pores 122 is too small, the adsorption capacity and the infiltrability of the electrolyte retention particles 100 for the electrolyte 340 are poor, and the liquid adsorption amount per unit particle is reduced. When the electrode 200 is assembled into the battery 300, the infiltrability of the electrode 200 for the electrolyte 340 is poor, thereby reducing the cycle performance and energy efficiency of the battery 300.

[0038] In some embodiments, each of the multiple sub-particles 121 has a particle size $D1$ satisfying: $50\text{ nm} \leq D1 \leq 90\text{ nm}$.

[0039] Specifically, the particle size $D1$ of the sub-particle 121 may be, but is not limited to, 50 nm, 55 nm, 58 nm, 60 nm, 64 nm, 68 nm, 70 nm, 72 nm, 75 nm, 78 nm, 80 nm, 82 nm, 84 nm, 86 nm, 88 nm, 90 nm, etc.

[0040] In this embodiment, when the particle size $D1$ of the sub-particle 121 satisfies: $50\text{ nm} \leq D1 \leq 90\text{ nm}$, the particle size $D1$ of the sub-particle 121 is in a reasonable range. Accordingly, when the outer diameter of the electrolyte retention particle 100 and the diameter of the core 110 are determined, the quantity of the sub-particles 121 of the shell 120 is in a reasonable range. When the multiple sub-particles 121 are jointed together to form the shell 120, the shell 120 has the higher structural strength. In addition, each sub-particle 121 has the capillary pores 122, so that the specific surface area of the electrolyte retention particle 100 can be effectively increased due to the sub-particle 121 with the capillary pores 122. The multiple capillary pores 122 have good adsorption performance for the electrolyte 340, so that the electrolyte retention particle 100 has the good electrolyte retention capacity, which is beneficial to improving the infiltrability of the electrode 200 for the electrolyte 340. When the particle size $D1$ of the sub-particle 121 is too large, the connections between the multiple sub-particles 121 are weakened, so that structural strength of the shell 120 is weaker, and then structural strength of the electrolyte retention particle 100 is reduced. When the electrode 200 is assembled into the battery 300, the electrolyte retention particle 100 with a weaker structural strength is prone to deformation or fracturing during charging and discharging of the battery 300, and then the overall structural strength of the electrode 200 is reduced. Moreover, the electrolyte 340 is unable to be effectively fixed and maintained, so that the infiltrability of the electrode 200 for the electrolyte 340 is reduced, thereby reducing the cycle performance and energy

efficiency of the battery 300. When the particle size D1 of the sub-particle 121 is too small, correspondingly, the radial dimension of the capillary pore 122 of the sub-particle 121 is too small, so that it is difficult for the capillary pore 122 to adsorb the electrolyte 340, thereby reducing the electrolyte retention effect of the electrolyte retention particle 100. When the electrode 200 is assembled into the battery 300, the infiltrability of the electrode 200 for the electrolyte 340 is poor, and the interface impedance between the electrode 200 and the electrolyte 340 is increased, so that the cycle performance of the battery 300 is poor and energy efficiency of the battery 300 is low.

[0041] In some embodiments, the core 110 has a diameter/2. Each of the electrolyte retention particles 100 has an outer diameter D3. D2 and D3 satisfy: $1:3 \leq D2:D3 \leq 1:1.2$.

[0042] In particular, the value of D2:D3 may be, but is not limited to, 1:3, 1:2.9, 1:2.7, 1:2.5, 1:2.4, 1:2.2, 1:2.0, 1:1.8, 1:1.7, 1:1.6, 1:1.5, 1:1.4, 1:1.3, 1:1.2, etc.

[0043] In this embodiment, when a ratio of the diameter D2 of the core 110 to the outer diameter D3 of the electrolyte retention particle 100 satisfies: $1:3 \leq D2:D3 \leq 1:1.2$, both the diameter D2 of the core 110 and the outer diameter D3 of the electrolyte retention particle 100 are in a reasonable range, in other words, both the diameter of the core 110 and the thickness of the shell 120 are in a reasonable range. The capillary pores 122 of the shell 120 can effectively adsorb the electrolyte 340, and the core 110 has a large enough volume to effectively store the electrolyte 340, so that the liquid adsorption amount per unit particle is high. When the electrode 200 is assembled into the battery 300, the electrode 200 has a good infiltrability for the electrolyte 340, thereby reducing the impedance at which the active ions in the electrolyte 340 react with active particles 500 of the electrode 200, so that the battery 300 has a high cycle performance and energy efficiency. When the value of D2:D3 is too large, the diameter D2 of the core 110 is too large, or the outer diameter D3 of the electrolyte retention particles 100 is too small, in other words, in the electrolyte retention particle 100, the diameter of the core 110 is too large, and the thickness of the shell 120 is too small. If the diameter of the core 110 is too large and the thickness of the shell 120 is too small, correspondingly, the quantity of the capillary pores 122 of the shell 120 is small, so that the adsorption amount of the electrolyte 340 by the shell 120 is reduced, and then the amount of the electrolyte 340 stored in the core 110 is reduced, thereby reducing the liquid adsorption amount per unit particle. When the electrolyte retention particle 100 is applied to the electrode 200 assembled into the battery 300, the electrode 200 has poor infiltrability for the electrolyte 340, thereby reducing the cycle performance and energy efficiency of the battery 300. When the value of D2:D3 is too small, the diameter D2 of the core 110 is too small, or the outer diameter D3 of the electrolyte retention particles 100 is too large, in other words, in the electrolyte retention particle 100, the diameter of the core 110 is too small, and the thickness of the shell 120 is too large. In the electrolyte retention particle 100 provided in this embodiment, after the capillary pores 122 of the shell 120 adsorb the electrolyte 340, since the volume of the core 110 is too small, it is difficult for the core 110 to store the electrolyte 340 effectively, and then the electrolyte 340 adsorbed by the capillary pores 122 overflows out of the electrolyte retention particle 100, thereby reducing the liquid adsorption amount per unit particle. When the electrode 200 is assembled into the

battery 300, the electrode 200 and the electrolyte 340 have poor infiltrability, thereby reducing the cycle performance and energy efficiency of the battery 300.

[0044] In some embodiments, the diameter D2 of the core 110 satisfies: $80 \text{ nm} \leq D2 \leq 300 \text{ nm}$.

[0045] Specifically, the diameter D2 of the core 110 may be, but is not limited to, 80 nm, 85 nm, 90 nm, 100 nm, 120 nm, 140 nm, 150 nm, 160 nm, 180 nm, 190 nm, 200 nm, 210 nm, 220 nm, 230 nm, 250 nm, 270 nm, 280 nm, 290 nm, 300 nm, etc.

[0046] In this embodiment, when the diameter D2 of the core 110 satisfies: $80 \text{ nm} \leq D2 \leq 300 \text{ nm}$, the diameter D2 of the core 110 is in a reasonable range, and in order to ensure the electrolyte retention effect of the electrolyte retention particle 100, the thickness of the shell 120 is also in a reasonable range. When the capillary pores 122 of the shell 120 adsorb the electrolyte 340, the electrolyte 340 can be effectively stored in the core 110, thereby greatly improving the electrolyte retention performance of the electrolyte retention particle 100. When the electrode 200 is assembled into the battery 300, the electrode 200 has a high infiltrability for the electrolyte 340, and the battery 300 has a good cycle performance. In addition, it is also possible to avoid increasing difficulty in compacting the electrode 200 due to an excessively large particle size of the electrolyte retention particle 100, so that the electrode 200 has a relatively high compaction density and energy density when the electrode 200 is applied to the battery 300. When the diameter of the core 110 is too large, in order to ensure the electrolyte retention effect of the electrolyte retention particle 100, the thickness of the shell 120 is also large enough, so that the outer diameter of the electrolyte retention particle 100 is too large, in other words, the particle size of the electrolyte retention particle 100 is too large. When the electrolyte retention particle 100 is applied to the electrode 200, there is a large gap between the electrolyte retention particle 100 with an excessively large particle size and the active particle 500, so that the compaction density of the electrode 200 is small, and the energy density of the electrode 200 is reduced. When the diameter of the core 110 is excessively small, after the shell 120 adsorbs the electrolyte 340, the core 110 is insufficient to store the electrolyte 340, so that the electrolyte 340 overflows out of the electrolyte retention particle 100, thereby reducing the electrolyte retention performance of the electrolyte retention particle 100. When the electrolyte retention particle 100 is applied to the electrode 200 assembled into the battery 300, the electrodes 200 have poor infiltrability for the electrolyte 340, resulting in poor cycle performance of the battery 300.

[0047] In some embodiments, the core 110 is a polymer synthesized from one or more selected from a group consisting of acrylic, acrylate, acrylamide, acrylonitrile, butadiene, styrene, and vinylidene fluoride.

[0048] In this embodiment, the core 110 is a polymer synthesized from monomers. The core 110 is the polymer synthesized from one or more selected from a group consisting of acrylic, acrylate, acrylamide, acrylonitrile, butadiene, styrene, and vinylidene fluoride. On one hand, the core 110 has a good adhesive performance, ensuring strong adhesion between the core 110 and the shell 120. In addition, when the electrolyte retention particle 100 is applied to the electrode 200, the electrolyte retention particle 100 also has a good adhesive performance with the active particle 500. On the other hand, the material selected for the core 110 has

low reactivity. When the electrolyte retention particle **100** is applied to the electrode **200** assembled into the battery **300**, a side reaction between the electrolyte retention particle **100** and the active particle **500** or the electrolyte **340** can be avoided, which helps to ensure the electrolyte retention performance of the electrolyte retention particle **100**. Consequently, the infiltrability of the electrode **200** for the electrolyte **340** is improved, and the cycle performance and energy efficiency of the battery **300** are improved.

[0049] Alternatively, the core **110** is selected from one of styrene butadiene rubber (SBR), polyacrylic acid (PAA), polyacrylonitrile (PAN), and polyvinylidene fluoride (PVDF).

[0050] In some embodiments, a material of the shell **120** is selected from one of silane, silicon dioxide, silicate ester, and aluminium oxide.

[0051] It may be understood that, the shell **120** is made of an inorganic material.

[0052] In this embodiment, the material of the shell **120** is selected from one of silane, silicon dioxide, silicate ester, and aluminium oxide. Compared to the shell **120** made of a polymer with poor oxidation resistance, since double bonds are absent from silane, silicon dioxide, silicate ester, and aluminium oxide, the shell **120** has high oxidation resistance. When the electrolyte retention particle **100** is applied to the electrode **200** and the electrode **200** is the positive electrode **310**, the electrolyte retention particle **100** is less prone to oxidation under high voltage. Therefore, the electrolyte retention particle **100** provided in the present disclosure may be for the negative electrode **330** and may be for the positive electrode **310**, thereby improving applicability of the electrolyte retention particle **100**.

[0053] Alternatively, in some embodiments, the capillary pores **122** of the sub-particle **121** of the shell **120** is prepared as follows. One of silane, silicon dioxide, silicate ester, and aluminium oxide is selected and mixed with a porous former, so that the shell **120** formed has multiple sub-particles **121**. The porous former is then volatilized by methods such as high temperature to obtain capillary pores **122** of the sub-particles **121**. Specifically, one of silane, silicon dioxide, silicate ester, and aluminium oxide is placed in an organic solvent (e.g., alkyl esters, cyclohexane), and then added them to an aqueous solution dissolved with alkyl ammonium cationic emulsifiers. The mixture is heated and stirred to allow the reaction to proceed. By a two-phase method, the selected one of silane, silicon dioxide, silicate ester, and aluminium oxide can be continuously precipitated and transferred into an aqueous phase. Finally, the organic solvent remaining on the surface of the sub-particle **121** is dried at a high temperature to form the sub-particle **121** having the capillary pores **122**.

[0054] In some embodiments, each of the electrolyte retention particles **100** has a specific surface area ranging from $85 \text{ m}^2/\text{g}$ to $350 \text{ m}^2/\text{g}$.

[0055] Specifically, the specific surface area of the electrolyte retention particle **100** may be, but is not limited to, $85 \text{ m}^2/\text{g}$, $88 \text{ m}^2/\text{g}$, $100 \text{ m}^2/\text{g}$, $120 \text{ m}^2/\text{g}$, $125 \text{ m}^2/\text{g}$, $150 \text{ m}^2/\text{g}$, $155 \text{ m}^2/\text{g}$, $188 \text{ m}^2/\text{g}$, $190 \text{ m}^2/\text{g}$, $200 \text{ m}^2/\text{g}$, $205 \text{ m}^2/\text{g}$, $235 \text{ m}^2/\text{g}$, $250 \text{ m}^2/\text{g}$, $265 \text{ m}^2/\text{g}$, $280 \text{ m}^2/\text{g}$, $295 \text{ m}^2/\text{g}$, $300 \text{ m}^2/\text{g}$, $310 \text{ m}^2/\text{g}$, $325 \text{ m}^2/\text{g}$, $335 \text{ m}^2/\text{g}$, $350 \text{ m}^2/\text{g}$, etc.

[0056] In this embodiment, when the specific surface area of the electrolyte retention particle **100** ranges from $85 \text{ m}^2/\text{g}$ to $350 \text{ m}^2/\text{g}$, the specific surface area of the electrolyte retention particle **100** is in a reasonable range. It may be

understood that, the capillary pores **122** of the sub-particle **121** greatly increase the specific surface area of the electrolyte retention particle **100**, so that the contact area between the electrolyte retention particle **100** and the electrolyte **340** is increased. After the pore **122** of the electrolyte retention particle **100** absorbs the electrolyte **340**, the electrolyte **340** can be stored in the core **110**, and the electrolyte retention particles **100** have a relatively strong electrolyte retention performance. When the electrolyte retention particle **100** is applied to the electrode **200** assembled into the battery **300**, the electrode **200** has good infiltrability for the electrolyte **340**, and thus the battery **300** has better cycle performance and higher energy efficiency. When the specific surface area of the electrolyte retention particle **100** is too large, correspondingly, the capillary pores **122** occupy a large area on the surface of the shell **120**, so that the structural strength of the shell **120** is weak. When the electrolyte retention particle **100** is applied to the electrode **200** assembled into the battery **300**, the electrolyte retention particle **100** with a weak structural strength is prone to deformation or fracturing during charging and discharging of the battery **300**, so that the overall structural strength of the electrode **200** is reduced. In addition, the electrolyte **340** is unable to be effectively fixed and retained, and the infiltrability of the electrode **200** for the electrolyte **340** is reduced, thereby reducing the cycle performance and energy efficiency of the battery **300**. When the specific surface area of the electrolyte retention particle **100** is too small, correspondingly, the capillary pores **122** occupy a small area on the surface of the shell **120**, so that the effective contact area between the electrolyte retention particle **100** and the electrolyte **340** is small, thereby reducing the adsorption capacity and infiltrability of the electrolyte retention particle **100** for the electrolyte **340**, and reducing the liquid adsorption amount per unit particle. When the electrolyte retention particle **100** is applied to the electrode **200** assembled into the battery **300**, the electrode **200** has poor infiltrability for the electrolyte **340**, thereby reducing cycle performance and energy efficiency of the battery **300**.

[0057] As illustrated in FIG. 5, a battery **300** is provided in the present disclosure. The battery **300** includes a positive electrode **310**, a separator **320**, a negative electrode **330**, and an electrolyte **340**. The separator **320** is disposed at one side of the positive electrode **310**. The negative electrode **330** is disposed at one side of the separator **320** positioned facing away from the positive electrode **310**. At least one of the positive electrode **310** or the negative electrode **330** is the electrode **200** provided in the present disclosure. At least a part of the positive electrode **310** and at least a part of the negative electrode **330** are immersed in the electrolyte **340**.

[0058] It may be understood that, the positive electrode **310**, the separator **320**, and the negative electrode **330** are stacked in sequence.

[0059] It may be understood that, in this embodiment, the positive electrode **310** is the electrode **200** provided in the present disclosure. In other embodiments, the negative electrode **330** is the electrode **200** provided in the present disclosure. In still other embodiments, each of the positive electrode **310** and the negative electrode **330** is the electrode **200** provided in the present disclosure.

[0060] In this embodiment, when at least one of the positive electrode **310** or the negative electrode **330** is the electrode **200** provided in the present disclosure, the electrode **200** has good infiltrability for the electrolyte **340**,

which helps to reduce the impedance at which the active ions in the electrolyte **340** react with the active particles **500** in the electrode **200**, so that the battery **300** has better cycle performance and higher energy efficiency.

[0061] Alternatively, the battery **300** may be, but is not limited to, a lithium ion battery, a sodium ion battery, or the like.

[0062] Alternatively, the battery **300** may be one of a cylindrical battery, a prismatic battery, a pouch battery, or the like.

[0063] Alternatively, the separator **320** is made of at least one of glass fiber, non-woven fabric, polyethylene, polypropylene, polyvinylidene fluoride, etc. In some embodiments, the separator **320** is a monolayer film. In other embodiments, the separator **320** is a multilayer composite film. When the separator **320** is a multilayer composite film, materials of various layers may be the same or different.

[0064] Alternatively, the separator **320** has a thickness ranging from 14 μm to 18 μm . Specifically, the thickness of the separator **320** may be, but is not limited to, 14 μm , 14.5 μm , 15 μm , 15.5 μm , 16 μm , 16.5 μm , 17 μm , 17.5 μm , 18 μm , etc.

[0065] Alternatively, the current collector layer **210** of the positive electrode **310** is selected from aluminum foil. The current collector layer **210** of the positive electrode **310** has good electrical conductivity and high mechanical strength.

[0066] Alternatively, the active layer **220** of the positive electrode **310** includes the active particles. The active particles of the positive electrode **310** are selected from lithium transition-metal oxide particles and modified particles thereof. The modified particles refer to the lithium transition-metal oxide particles that undergo doping modification and/or coating modification.

[0067] Preferably, the active particles of the positive electrode **310** are selected from one or more of lithium cobalt oxide, lithium nickel oxide, lithium manganese oxide, lithium nickel manganese oxide, lithium nickel cobalt manganese oxide, and lithium nickel cobalt aluminum oxide.

[0068] Alternatively, the active layer **220** of the positive electrode **310** further includes a positive adhesive agent and a positive conductive agent. The positive adhesive agent is configured to adhere various components (e.g., at least two of the active particle, the positive conductive agent, and the electrolyte retention particle **100**) in the active layer **220**, so as to improve overall performance of the positive electrode **310**. The positive conductive agent can improve the conductivity of the active layer **220**.

[0069] Alternatively, the positive adhesive agent is selected from at least one of polyvinylidene fluoride, polytetrafluoroethylene, etc.

[0070] Alternatively, the positive conductive agent is selected from at least one of graphite, superconducting carbon, acetylene black, carbon black, ketjen black, carbon dot, carbon nanotube, graphene, carbon nanofiber, etc.

[0071] Alternatively, the current collector layer **210** of the negative electrode **330** is selected from copper foil. The current collector layer **210** of the negative electrode **330** has good electrical conductivity and high mechanical strength.

[0072] Alternatively, the active particles of the negative electrode **330** are selected from one or more of graphite, meso-carbon micro-bead, hard carbon, and soft carbon, with graphite being preferred. The graphite may be selected from one or more of synthetic graphite and natural graphite.

[0073] Alternatively, the active layer **220** of the negative electrode **330** further includes a negative adhesive agent, a negative conductive agent, and a thickener. The negative adhesive agent is configured to adhere components (i.e., at least two of the active particle, the negative conductive agent, the electrolyte retention particle **100**, and the thickener) in the active layer **220**, so as to improve overall performance of the negative electrode **330**. The negative conductive agent can improve electrical conductivity of the active layer **220** of the negative electrode **330**. The thickener can improve adhesion of the active layer **220** of the negative electrode **330**.

[0074] Alternatively, the negative conductive agent is selected from at least one of acetylene black, conductive carbon black, carbon nanotube, carbon fiber, graphene, etc.

[0075] Alternatively, the negative adhesive agent is selected from one or more of polyacrylic acid (PAA), sodium polyacrylate (PAAS), polyvinyl alcohol (PVA), styrene-butadiene rubber (SBR), carboxymethyl cellulose (CMC), sodium alginate, polymethyl methacrylate (PMA), carboxymethyl chitosan (CMCS), etc.

[0076] Alternatively, the thickener is selected from at least one of sodium carboxymethyl cellulose (Na CMC), polyacrylamide (PAM), PMA.

[0077] Alternatively, the electrolyte **340** includes an electrolyte salt and a solvent. The solvent is used for dissolving the electrolyte salt, so that the electrolyte salt releases active ions.

[0078] Alternatively, the electrolyte salt is selected from one or more of lithium hexafluorophosphate (LiPF_6), lithium tetrafluoroborate (LiBF_4), lithium perchlorate (LiClO_4), lithium hexafluoroarsenate (LiAsF_6), lithium bis(trifluoromethanesulfonyl)imide (LiFSI), lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), lithium trifluoromethanesulfonate (LiTFS), lithium difluoroxyborate (LiDFOB), lithium bis(oxalate)borate (LiBOB), lithium difluorophosphate (LiPO_2F_2), lithium difluoro(oxalato)phosphate (LiDFOP), and lithium tetrafluoro(oxalato)phosphate (LiTFOP).

[0079] Alternatively, the solvent is selected from one or more of ethylene carbonate (EC), propylene carbonate (PC), ethyl methyl carbonate (EMC), diethyl carbonate (DEC), dimethyl carbonate (DMC), dipropyl carbonate (DPC), methyl propyl carbonate (MPC), ethyl propyl carbonate (EPC), butylene carbonate (BC), fluoroethylene carbonate (FEC), methyl formate (MF), methyl acetate (MA), ethyl acetate (EA), propyl acetate (PA), methyl propionate (MP), ethyl propionate (EP), propyl propionate (PP), methyl butyrate (MB), ethyl butyrate (EB), γ -butyrolactone (GBL), sulfolane (SF), dimethyl sulfone (MSM), ethyl methyl sulfone (EMS), and diethyl sulfone (ESE).

[0080] Alternatively, the electrolyte **340** further includes an additive. The additive is dissolved in the solvent. The additive may be, but is not limited to, a positive film-forming additive or a negative film-forming additive. The additive may also be an additive for improving performance of the battery **300**. Alternatively, the additive may be an additive for improving overcharge resistance of the battery **300**, an additive for improving high temperature resistance of the battery **300**, or an additive for improving low temperature resistance of the battery **300**.

[0081] In some embodiments, the positive electrode **310** is the electrode **200** provided in the present disclosure. A mass

fraction A1 of the electrolyte retention particles 100 in the active layer 220 of the positive electrode 310 satisfies: $0.4\% \leq A1 \leq 1.2\%$.

[0082] Specifically, the mass fraction A1 of the electrolyte retention particles 100 may be, but is not limited to, 0.4%, 0.45%, 0.48%, 0.5%, 0.52%, 0.58%, 0.6%, 0.65%, 0.7%, 0.75%, 0.8%, 0.9%, 0.95%, 1%, 1.05%, 1.1%, 1.15%, 1.2%, etc.

[0083] It may be understood that, in the active layer 220 of the positive electrode 310, the mass fraction of the electrolyte retention particles 100 is a ratio of the mass of the electrolyte retention particles 100 to the mass of the active layer 220 of the positive electrode 310.

[0084] In this embodiment, when the mass fraction A1 of the electrolyte retention particles 100 in the active layer 220 of the positive electrode 310 satisfies $0.4\% \leq A1 \leq 1.2\%$, the mass fraction of the electrolyte retention particles 100 is in a reasonable range. On one hand, the electrolyte retention particles 100 have better electrolyte retention performance, can effectively improve the infiltrability of the positive electrode 310 for the electrolyte 340, and reduce the impedance of active ions in the electrolyte 340 at the interface between the positive electrode 310 and the electrolyte 340, so that the battery 300 has better cycle performance. On the other hand, the mass fraction of the active particles of the positive electrode 310 is also in a reasonable range, so that the positive electrode 310 has a higher energy density, which helps to prolong the cycle life of the battery 300. When the mass fraction of the electrolyte retention particles 100 is too large, that is, when the mass fraction of the electrolyte retention particles 100 in the active layer 220 of the positive electrode 310 is too large, the mass fraction of the active particles is too small, thereby reducing the energy density of the positive electrode 310. In addition, when the positive electrode 310 is applied to the battery 300, the cycle life of the battery 300 is shortened. When the mass fraction of the electrolyte retention particles 100 is too small, that is, when the mass fraction of the electrolyte retention particles 100 in the active layer 220 of the positive electrode 310 is too small, it is difficult for the electrolyte retention particles 100 to improve the infiltrability of the positive electrode 310 for the electrolyte 340. When the positive electrode 310 is applied to the battery 300, the impedance of active ions in the electrolyte 340 at the interface between the electrolyte 340 and the positive electrode 310 is too large, so that the cycle performance of the battery 300 is poor and the energy efficiency of the battery 300 is low.

[0085] In some embodiments, the negative electrode 330 is the electrode 200 provided in the present disclosure. A mass fraction A2 of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 satisfies: $0.2\% \leq A2 \leq 0.8\%$.

[0086] Specifically, the mass fraction A2 of the electrolyte retention particles 100 may be, but is not limited to, 0.2%, 0.25%, 0.3%, 0.35%, 0.4%, 0.45%, 0.48%, 0.5%, 0.52%, 0.55%, 0.6%, 0.65%, 0.7%, 0.75%, 0.8%, etc.

[0087] It may be understood that, in the active layer 220 of the negative electrode 330, the mass fraction of the electrolyte retention particles 100 is a ratio of the mass of the electrolyte retention particles 100 to the mass of the active layer 220 of the negative electrode 330.

[0088] In this embodiment, when the mass fraction A2 of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 satisfies $0.4\% \leq A2 \leq 1.2\%$, the

mass fraction of the electrolyte retention particles 100 is in a reasonable range. On one hand, the electrolyte retention particles 100 have better electrolyte retention performance, can effectively improve the infiltrability of the negative electrode 330 for the electrolyte 340, and reduce the impedance of active ions in the electrolyte 340 at the interface between the negative electrode 330 and the electrolyte 340, so that the battery 300 has better cycle performance. On the other hand, the mass fraction of the active particles of the negative electrode 330 is also in a reasonable range, so that the negative electrode 330 has a higher energy density, which helps to prolong the cycle life of the battery 300. When the mass fraction of the electrolyte retention particles 100 is too large, that is, when the mass fraction of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 is too large, the mass fraction of the active particles is too small, thereby reducing the energy density of the negative electrode 330. In addition, when the negative electrode 330 is applied to the battery 300, the cycle life of the battery 300 is shortened. When the mass fraction of the electrolyte retention particles 100 is too small, that is, when the mass fraction of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 is too small, it is difficult for the electrolyte retention particles 100 to improve the infiltrability of the negative electrode 330 for the electrolyte 340. When the negative electrode 330 is applied to the battery 300, the impedance of active ions in the electrolyte 340 at the interface between the electrolyte 340 and the negative electrode 330 is too large, so that the cycle performance of the battery 300 is poor and the energy efficiency of the battery 300 is low.

[0089] The technical solutions of the present disclosure will be further described below through multiple embodiments.

Examples 1 to 8, and Comparative Example

1. Preparation of Electrolyte Retention Particles 100

[0090] First, sub-particles 121 are formed. Specifically, one of silane, silicon dioxide, silicate ester, and aluminum oxide is placed in an organic solvent (e.g., alkyl esters or cyclohexane). The organic solvent serves as a pore-forming agent. The mixture is added to an aqueous solution dissolved with alkyl ammonium cationic emulsifiers, followed by heating and stirring to allow the reaction to proceed. By a two-phase method, the selected one of silane, silicon dioxide, silicate ester, and aluminum oxide can be continuously precipitated and transferred into an aqueous phase. Finally, the organic solvent remaining on the surface of the sub-particle 121 is dried at a high temperature to form the sub-particles 121 having capillary pores 122. The radial dimension d of each capillary pore 122 satisfies: $2 \text{ nm} \leq d \leq 30 \text{ nm}$. A distribution density a of each capillary pore 122 satisfies: $0.4 \text{ cm}^3/\text{g} \leq a \leq 3 \text{ cm}^3/\text{g}$. The radial dimension d and distribution density a of the capillary pore 122 can be controlled by adjusting the type of organic solvent, the solubility of silane, silicon dioxide, silicate ester, and aluminum oxide in the organic solvent, and the concentration of silane, silicon dioxide, silicate ester, and aluminum oxide in the organic solvent.

[0091] Furthermore, a polymer synthesized from one or more selected from a group consisting of acrylic, acrylate, acrylamide, acrylonitrile, butadiene, styrene, and vinylidene fluoride is used as a material for the core 110. The sub-

particles **121** is mixed and stirred with the material of the core **110** for adsorption to form the electrolyte retention particles **100**. A ratio of the core **110** to the electrolyte retention particles **100** (D3) is controlled by the mass ratio of the sub-particles **121** to the material of the core **110**.

2. Preparation of the Electrode **200**

(1) Preparation of the Positive Electrode **310**

[0092] Active particles, electrolyte retention particles **100**, a positive conductive agent, and a positive adhesive agent, that are suitable for the active layer **220** of the positive electrode **310**, are dispersed in a solvent (e.g., N-methyl pyrrolidone) to form a uniform positive slurry. The positive slurry is coated onto the current collector layer **210** to form the active layer **220** of the positive electrode **310**. After drying, cold pressing, and other processes, the positive electrode **310** is obtained. The mass fraction A1 of the liquid-retaining particles **100** in the active layer **220** of the positive electrode **310**, as used in each of examples 1 to 8 and the comparative example, is shown in Table 1.

(2) Preparation of the Negative Electrode **330**

[0093] Active particles, electrolyte retention particles, a negative conductive agent, a negative adhesive agent, and a thickener, that are suitable for the active layer **220** of the negative electrode **330**, are dispersed in a solvent (e.g., deionized water) to form a uniform negative slurry. The negative slurry is coated onto the negative current collector to form the active layer **220** of the negative electrode **330**. After drying, cold pressing, and other processes, the nega-

until completely dissolved, and an additive is added. After thorough mixing, the electrolyte **340** is obtained.

4. Assembly of the Battery **300**

[0096] The positive electrode **310** and the separator **320** prepared in each of examples 1 to 8 and comparative example, and the negative electrode **330** prepared in each of examples 1 to 8 and comparative example are laminated in order, ensuring that the separator **320** is located between the positive electrode **310** and the negative electrode **330**, and then are wound to obtain a bare cell. The bare cell is placed in an outer package, dried, and then injected with the aforementioned electrolyte **340**. After undergoing processes such as standing, formation, encapsulation, etc., example batteries 1 to 8 and a comparative battery are obtained. The positive electrode **310** in example 1 and the negative electrode **330** in example 1 are provided in the example battery 1. The positive electrode **310** in example 2 and the negative electrode **330** in example 2 are provided in the example battery 2. The positive electrode **310** in example 3 and the negative electrode **330** in example 3 are provided in the example battery 3, and so on.

[0097] Table 1 below shows structural parameters of the positive electrode **310** and the negative electrode **330** in each of examples 1 to 8 and comparative example. Table 1 shows the mass fraction A1 of the electrolyte retention particles **100** in the active layer **220** of the negative electrode **330** in each of examples 1 to 8 and comparative examples, and the mass fraction A2 of the electrolyte retention particles **100** in the active layer **220** of the negative electrode **330** in each of examples 1 to 8 and comparative example.

TABLE 1

Structural parameters of the positive electrode 310 and the negative electrode 330 in each of examples 1 to 8 and comparative example		
Example and comparative example	Mass fraction A1 of electrolyte retention particles 100 in the active layer 220 of the positive electrode 310 (%)	Mass fraction A2 of electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 (%)
Example 1	0	0.2
Example 2	0	0.4
Example 3	0	0.6
Example 4	0	0.8
Example 5	0	1
Example 6	0.4	0
Example 7	0.8	0
Example 8	0.8	0.8
Comparative example	0	0

tive electrode **330** is obtained. The mass fraction A2 of the liquid-retaining particles **100** in the active layer **220** of the negative electrode **330**, as used in each of examples 1 to 8 and the comparative example, is shown in Table 1.

3. Preparation of the Separator **320** and the Electrolyte **340**

[0094] A polyethylene separator **320** having the thickness of 16 μm is selected.

[0095] In a glove box under an argon atmosphere with a moisture content of ≤ 1 ppm, ethylene carbonate, dimethyl carbonate, and ethyl methyl carbonate are mixed at a mass fraction of 1:1:1, a dried electrolyte salt (e.g., lithium hexafluorophosphate) is dissolved into a solvent, stirred

Performance Test of the Battery **300**

1. Test of the Infiltrability of the Electrode **200** for the Electrolyte **340**

[0098] At 25° C., a sample of the electrode **200** prepared in each of examples 1 to 8 and comparative example is cut to a fixed length of 14 cm and a fixed width of 1.5 cm. The sample is horizontally straightened, and then 2 mL of the electrolyte **340** is dropped onto the sample of the electrode **200** by using a syringe. After 3 minutes, the infiltration length of the electrode **200** is measured. It may be seen that the infiltration length refers to the length of part of the

sample of the electrode **200**, straightened horizontally, that is infiltrated by the electrolyte **340** in the straightened direction.

[0099] The electrode **200** includes the positive electrode **310** and the negative electrode **330**. The infiltration length of the positive electrode **310** for the electrolyte **340** and the infiltration length of the negative electrode **330** for the electrolyte **340** in each of examples 1 to 8 are shown in Table 2.

2. Test of Internal Resistance Performance of the Battery **300**

(1) Calculation of Charge DCR (Direct Current Internal Resistance) at 50% SOC (State of Charge)

[0100] The test temperature is 25° C. After formation, the example batteries 1 to 8 and the comparative battery are charged at a constant current of 0.5C until a voltage is 3.65V, then charged at a constant voltage of 3.65V until the current is 0.05C, and left to stand for 10 minutes. Subsequently, the example batteries 1 to 8 and the comparative battery are discharged at a constant current of 0.5C until the voltage is 2.5V, and left to stand for 10 minutes, where the capacity at this step was recorded as C₀. At 25° C., the example batteries 1 to 8 and the comparative battery are charged at a constant current of 1C for 30 minutes. At this time, the example batteries 1 to 8 and the comparative battery each are at 50% SOC. After being left to stand for 1 hour, the example batteries 1 to 8 and the comparative battery are discharged at 1.07C₀ for 30 seconds. After being left to stand for 40 seconds, V₁ of each of the example batteries 1 to 8 and the comparative battery is recorded. The example batteries 1 to 8 and the comparative battery are charged at 1.07C₀ for 30 s, and then V₂ of each of the example batteries 1 to 8 and the comparative battery is recorded. After being left to stand for 10 minutes, the DCR of each of the example batteries 1 to 8 and the comparative battery at 50% SOC is calculated by $DCR=(V_2-V_1)/1.07C_0$.

(2) Calculation of Discharge DCR at 50% SOC

[0101] The test temperature is 25° C. After formation, the example batteries 1 to 8 and the comparative battery are charged at a constant current of 0.5C₀ until a voltage is 3.65V, then charged at a constant voltage of 3.65V until the current is 0.05C, and left to stand for 10 minutes. Subsequently, the example batteries 1 to 8 and the comparative battery are discharged at a constant current of 1C for 30 minutes. At this time, the example batteries 1 to 8 and the comparative battery each are at 50% SOC. After being left

to stand for 1 hour, V₃ of each of the example batteries 1 to 8 and the comparative battery is recorded. Then, the example batteries 1 to 8 and the comparative battery are discharged at 1.07C₀ for 30 s, and after being left to stand for 40 s, V₄ of each of the example batteries 1 to 8 and the comparative battery is recorded. After being left to stand for 10 minutes, the DCR' of each of the example batteries 1 to 8 and the comparative battery at 50% SOC is calculated by $DCR'=(V_3-V_4)/1.07C_0$.

[0102] The discharge DCR at 50% SOC refers to the direct current internal resistance measured during the discharge operation when the battery **300** is at 50% SOC. The charge DCR at 50% SOC refers to the direct current internal resistance measured during the charge operation when the battery **300** is at 50% SOC. The discharge DCR and the charge DCR of each of the example batteries 1 to 8 and the comparative battery at 50% SOC are shown in Table 2.

3. Test of Cycle Performance of the Battery **300**

[0103] After being left to stand at 25° C. for 1 hour, the example batteries 1 to 8 and the comparative battery are charged at 0.5P until a voltage is 3.65V, and then discharged at 0.5P until the voltage is 2.5V. An initial discharge capacity C₁ of the battery **300** and an initial discharge energy E₁ of the battery **300** are recorded. Each of the example batteries 1 to 8 and the comparative battery is cycled through three charge-discharge cycles, and a discharge energy E₃ and a charge energy E'₃ of each of the example batteries 1 to 8 and the comparative battery are recorded. Therefore, an energy efficiency E of each of the example batteries 1 to 8 and the comparative battery after being cycled three charge-discharge cycles can be calculated by $E=E_3/E'_3 \times 100\%$. Each of the example batteries 1 to 8 and the comparative battery is cycled through one charge-discharge cycle, and an initial discharge energy C₁ of each of the example batteries 1 to 8 and the comparative battery is recorded. Each of the example batteries 1 to 8 and the comparative battery is cycled through two hundred charge-discharge cycles, and a discharge capacity C₂₀₀ of each of the example batteries 1 to 8 and the comparative battery is recorded. Therefore, a capacity retention rate Cf of each of the example batteries 1 to 8 and the comparative battery can be calculated by $Cf=C_{200}/C_1 \times 100\%$.

[0104] The energy efficiency E and the capacity retention rate Cf after 500 cycles of each of the example batteries 1 to 8 and the comparative battery are shown in Table 2.

[0105] Table 2 below shows performance parameters of each of the example batteries 1 to 8 and the comparative battery.

TABLE 2

Performance parameters of the example batteries 1 to 8 and the comparative battery						
Example batteries and comparative battery	Infiltration length of positive electrode for electrolyte (cm)	Infiltration length of negative electrode for electrolyte (cm)	Discharge DCR at 50% SOC (mΩ)	Charge DCR at 50% SOC (mΩ)	Cycle energy efficiency after three cycles (%)	Cycle capacity retention rate after two hundred cycles (%)
Example 1	1.2	2.8	25.034	29.998	93.18	97.6
Example 2	1.2	3.2	24.567	29.245	93.23	97.9
Example 3	1.2	3.5	24.121	28.897	93.30	98.2
Example 4	1.2	4.1	23.564	27.985	93.40	98.4
Example 5	1.2	4.2	23.240	27.775	93.45	99.1
Example 6	1.5	2.5	24.778	29.357	93.20	98.2

TABLE 2-continued

Performance parameters of the example batteries 1 to 8 and the comparative battery						
Example batteries and comparative battery	Infiltration length of positive electrode for electrolyte (cm)	Infiltration length of negative electrode for electrolyte (cm)	Discharge DCR at 50% SOC (mΩ)	Charge DCR at 50% SOC (mΩ)	Cycle energy efficiency after three cycles (%)	Cycle capacity retention rate after two hundred cycles (%)
Example 7	2.0	2.5	23.988	28.130	93.37	99.2
Example 8	2.0	4.1	23.698	27.24	93.50	99.5
Comparative example	1.2	2.5	25.698	30.24	93.15	97.5

[0106] As illustrated in Table 1 and Table 2, the positive electrode 310 and the negative electrode 330 in comparative example 1 both do not include electrolyte retention particles 100. At 25° C., the infiltration length of the positive electrode 310 for the electrolyte 340 is 1.2 cm, and the infiltration length of the negative electrode 330 for the electrolyte 340 is 2.5 cm. The infiltration length of the electrode 200 for the electrolyte 340 can represent the infiltrability of the electrode 200 for the electrolyte 340. Specifically, when the infiltration length of the positive electrode 310 for the electrolyte 340 is longer, the infiltrability of the positive electrode 310 for the electrolyte 340 is better. When the infiltration length of the negative electrode 330 for the electrolyte 340 is longer, the infiltrability of the negative electrode 330 for the electrolyte 340 is better.

[0107] Referring to examples 1 to 5 and comparative example, the mass fraction A1 of the electrolyte retention particles 100 in the active layer 220 of the positive electrode 310 in each of examples 1 to 5 is 0. In other words, the active layer 220 of the positive electrode 310 in each of examples 1 to 5 does not include the electrolyte retention particles 100. The mass fraction A2 of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 in each of examples 1 to 4 satisfies: $0.2\% \leq A2 \leq 0.8\%$. The mass fraction A2 of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 in example 5 is greater than 0.8. It may be seen from performance parameters of the example batteries 1 to 5 that, the infiltration length of the negative electrode 330 in each of the example batteries 1 to 5 for the electrolyte 340 is longer than the infiltration length of the negative electrode 330 in the comparative battery for the electrolyte 340. The discharge DCR of each of example batteries 1 to 5 at 50% SOC is less than the discharge DCR of the comparative battery at 50% SOC. The charge DCR of each of example batteries 1 to 5 at 50% SOC is less than the charge DCR of the comparative battery at 50% SOC. In addition, the cycle energy efficiency after three cycles of each of the example batteries 1 to 5 is higher than the cycle energy efficiency after three cycles of the comparative battery. The cycle capacity retention rate after two hundred cycles of each of the example batteries 1 to 5 is higher than the cycle capacity retention rate after two hundred cycles of the comparative battery. The reasons lie below. The negative electrode 330 in each of examples 1 to 5 includes the electrolyte retention particles 100, each of the electrolyte retention particles 100 includes the core 110 and the shell 120, the shell 120 has multiple capillary pores 122, and each capillary pore 122 may be used to absorb the electrolyte 340 and the electrolyte 340 is stored in the core 110, so that the electrolyte retention particle 100 has good

electrolyte retention performance. When the electrolyte retention particles 100 are applied to the negative electrode 330, the infiltrability of the negative electrode 330 for the electrolyte 340 is improved, and then the infiltration length of the negative electrode 330 for the electrolyte 340 is increased. Moreover, when the infiltrability of the negative electrode 330 for the electrolyte 340 is better, the effective contact area between the electrolyte 340 and the negative electrode 330 is larger, so that the impedance of the active ions in the battery 300 at the interface between the negative electrode 330 and the electrolyte 340 is reduced, and then the discharge DCR of each of the example batteries 1 to 5 at 50% SOC is less than the discharge DCR of the comparative battery at 50% SOC, and the charge DCR of each of the example batteries 1 to 5 at 50% SOC is less than the charge DCR of the comparative battery at 50% SOC. It is more conducive to charging and discharging processes of the battery 300, finally resulting in relatively high cycle energy efficiency and capacity retention rate of each of the example batteries 1 to 5, and better cycle performance of each of the example batteries 1 to 5 than the comparative battery. Further, as the mass fraction of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 is increased, the infiltration length of the negative electrode 330 for the electrolyte 340 is continuously increased. When A2 is greater than 0.8%, it is also difficult to increase the infiltration length of the negative electrode 330 for the electrolyte 340 even if the mass fraction of the electrolyte retention particles 100 is continuously increased. Therefore, in order to save the raw material cost of the negative electrode 330, it is necessary to avoid excessive electrolyte retention particles 100. When A2 satisfies: $0.2\% \leq A2 \leq 0.8\%$, not only can the negative electrode 330 have high infiltrability for the electrolyte 340, but also the raw material cost of the negative electrode 330 can be effectively controlled.

[0108] Referring to example 6, example 7, and comparative example, the mass fraction A1 of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 in each of example 6 and example 7 is 0. In other words, the active layer 220 of the positive electrode 310 in each of example 6 and example 7 does not include the electrolyte retention particles 100. The mass fraction A1 of the electrolyte retention particles 100 in the active layer 220 of the negative electrode 330 in each of example 6 and example 7 satisfies: $0.4\% \leq A1 \leq 1.2\%$. It may be seen from performance parameters of the example battery 6 and the example battery 7 that, the infiltration length of the positive electrode 310 in each of the example battery 6 and the example battery 7 for the electrolyte 340 is longer than the

infiltration length of the positive electrode **310** in the comparative battery for the electrolyte **340**. The discharge DCR of each of the example battery 6 and the example battery 7 at 50% SOC is less than the discharge DCR of the comparative battery at 50% SOC. The charge DCR of each of the example battery 6 and the example battery 7 at 50% SOC is less than the charge DCR of the comparative battery at 50% SOC. In addition, the cycle energy efficiency after three cycles of each of the example battery 6 and the example battery 7 is higher than the cycle energy efficiency after three cycles of the comparative battery. The cycle capacity retention rate after two hundred cycles of each of the example battery 6 and the example battery 7 is higher than the cycle capacity retention rate after two hundred cycles of the comparative battery. The reasons lie below. The positive electrode **310** in each of the example battery 6 and the example battery 7 includes the electrolyte retention particles **100**, each of the electrolyte retention particles **100** includes the core **110** and the shell **120**, the shell **120** has multiple capillary pores **122**, and each capillary pore **122** may be used to absorb the electrolyte **340** and the electrolyte **340** is stored in the core **110**, so that the electrolyte retention particle **100** has good electrolyte retention performance. When the electrolyte retention particles **100** are applied to the positive electrode **310**, the infiltrability of the positive electrode **310** for the electrolyte **340** is improved, and then the infiltration length of the positive electrode **310** for the electrolyte **340** is increased. Moreover, when the infiltrability of the positive electrode **310** for the electrolyte **340** is better, the effective contact area between the electrolyte **340** and the positive electrode **310** is larger, so that the impedance of the active ions in the battery **300** at the interface between the positive electrode **310** and the electrolyte **340** is reduced, and then the discharge DCR of each of the example battery 6 and the example battery 7 at 50% SOC is less than the discharge DCR of the comparative battery at 50% SOC, and the charge DCR of each of each of the example battery 6 and the example battery 7 at 50% SOC is less than the charge DCR of the comparative battery at 50% SOC. It is more conducive to charging and discharging processes of the battery **300**, finally resulting in relatively high cycle energy efficiency and capacity retention rate of each of the example battery 6 and the example battery 7, and better cycle performance of each of the example battery 6 and the example battery 7 than the comparative battery.

[0109] Further, referring to example 7 and example 8, the positive electrode **310** and the negative electrode **330** in example 8 both include the electrolyte retention particles **100**. Compared to an example in which either the positive electrode **310** or the negative electrode **330** is the electrode **200** provided in the present disclosure, the discharge DCR of the example battery 8 at 50% SOC is less than the discharge DCR of the example battery 7 at 50% SOC, the charge DCR of the example battery 8 at 50% SOC is less than the charge DCR of the example battery 7 at 50% SOC; and the cycle energy efficiency after three cycles of the example battery 8 is greater than the cycle energy efficiency after three cycles of the example battery 7, and the cycle capacity retention rate after two hundred cycles of the example battery 8 is greater than the cycle capacity retention rate after two hundred cycles of the example battery 7. The reasons lie below. Both the positive electrode **310** and the negative electrode **330** in the example battery 8 include the electrolyte retention particles **100**, the mass fraction A1 of the

electrolyte retention particles **100** in the active layer **220** of the positive electrode **310** and the mass fraction A2 of the electrolyte retention particles **100** in the active layer **220** of the negative electrode **330** are both in the reasonable range. The electrolyte retention particles **100** have good electrolyte retention performance, so that the positive electrode **310** and the negative electrode **330** both have good infiltrability for the electrolyte **340**. Therefore, the impedance of active ions in the battery **300** at the interface between the positive electrode **310** and the electrolyte **340** is greatly reduced, and the impedance of the active ions in the battery **300** at the interface between the negative electrode **330** and the electrolyte **340** is greatly reduced, so that the example battery 8 has higher cycle performance and energy efficiency.

[0110] As illustrated in FIG. 6 and FIG. 7, a power consuming device **400** is provided in the present disclosure. The power consuming device **400** includes a device body **410** and the battery **300** provided in the present disclosure. The battery **300** is configured to supply power to the device body **410**.

[0111] It may be understood that the battery **300** is electrically connected to the device body **410**.

[0112] In this embodiment, the battery **300** has good cycle performance and high energy efficiency, so that the battery **300** has a long service life. When the battery **300** is used to supply power to the device body **410**, the battery **300** can provide a stable power supply for the device body **410**, so that the device body **410** operates normally.

[0113] Alternatively, the power consuming device **400** in embodiments of the present disclosure may be, but is not limited to, a portable electronic device, such as a mobile phone, a tablet computer, a laptop, a desktop computer, a smart bracelet, a smart watch, an e-book reader, and a game console. The power consuming device **400** may also be transportation means such as an automobile, a truck, a car, a van, a bullet train, a high-speed train, and an electric bicycle. In addition, the power consuming device **400** may also be various household appliances, etc. The power consuming device **400** in the embodiment of the present disclosure in FIG. 7 is an energy storage battery cabinet.

[0114] It may be understood that the power consuming device **400** described in this implementation is only a form of the power consuming device **400** to which the battery **300** is applied, and should not be understood as a limitation to the power consuming device **400** provided in the present disclosure, nor should it be understood as a limitation to the battery **300** provided in various implementations of the present disclosure.

[0115] Reference in the present disclosure to an “embodiment” or “implementation” means that a particular feature, structure, or characteristic described in combination with the embodiments may be included in at least one embodiment of the present disclosure. The appearances of the phrase in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments mutually exclusive of the other embodiments. It is explicitly and implicitly understood by those of ordinary skill in the art that the embodiments described in the present disclosure may be combined with other embodiments. In addition, it should also be understood that, if there is no contradiction therebetween, the features, structures, or characteristics described in various embodiments of the present disclosure may be arbitrarily combined

to form another embodiment that does not depart from the spirit and scope of the technical solutions of the present disclosure.

[0116] Finally, it should be noted that the foregoing implementations are only intended for describing the technical solutions of the present disclosure but not for limiting the present disclosure. Although the present disclosure is described in detail with reference to the above preferred embodiments, a person of ordinary skill in the art should understand that modifications or equivalent replacements can be made to the technical solutions of the present disclosure without departing from the spirit and scope of the technical solutions of the present disclosure.

What is claimed is:

1. An electrode, comprising:
a current collector layer; and
an active layer, disposed on a surface of the current collector layer, wherein the active layer comprises active particles and electrolyte retention particles; and each of the electrolyte retention particles comprises a core and a shell, the core is coated with the shell, the shell comprises a plurality of sub-particles, and each of the plurality of sub-particles has capillary pores.
2. The electrode of claim 1, wherein each of the capillary pores has a radial dimension d satisfying: $2\text{ nm} \leq d \leq 30\text{ nm}$.
3. The electrode of claim 1, wherein in the shell, the capillary pores have a distribution density a satisfying: $0.4\text{ cm}^3/\text{g} \leq a \leq 3\text{ cm}^3/\text{g}$.
4. The electrode of claim 1, wherein each of the plurality of sub-particles has a particle size $D1$ satisfying: $50\text{ nm} \leq D1 \leq 90\text{ nm}$.
5. The electrode of claim 1, wherein the core has a diameter $D2$, and each of the electrolyte retention particles has an outer diameter $D3$, $D2$ and $D3$ satisfying: $1:3 \leq D2:D3 \leq 1:1.2$.
6. The electrode of claim 5, wherein the diameter $D2$ of the core satisfies: $80\text{ nm} \leq D2 \leq 300\text{ nm}$.
7. The electrode of claim 1, wherein the core is a polymer synthesized from one or more selected from a group consisting of acrylic, acrylate, acrylamide, acrylonitrile, butadiene, styrene, and vinylidene fluoride.
8. The electrode of claim 1, wherein a material of the shell is selected from one of silane, silicon dioxide, silicate ester, and aluminium oxide.
9. The electrode of claim 1, wherein each of the electrolyte retention particles has a specific surface area ranging from $85\text{ m}^2/\text{g}$ to $350\text{ m}^2/\text{g}$.
10. A battery, comprising:
a positive electrode;
a separator, disposed at one side of the positive electrode;
a negative electrode, disposed at one side of the separator positioned facing away from the positive electrode, wherein at least one of the positive electrode or the negative electrode is an electrode comprising:
a current collector layer; and
an active layer, disposed on a surface of the current collector layer, wherein the active layer comprises active particles and electrolyte retention particles; and each of the electrolyte retention particles comprises a core and a shell, the core is coated with the shell, the shell comprises a plurality of sub-particles, and each of the plurality of sub-particles has capillary pores; and

an electrolyte, wherein at least a part of the positive electrode and at least a part of the negative electrode are immersed in the electrolyte.

11. The battery of claim 10, wherein the positive electrode is the electrode, and a mass fraction $A1$ of the electrolyte retention particles in the active layer of the positive electrode satisfies: $0.4\% \leq A1 \leq 1.2\%$.

12. The battery of claim 10, wherein the negative electrode is the electrode, and a mass fraction $A2$ of the electrolyte retention particles in the active layer of the negative electrode satisfies: $0.2\% \leq A2 \leq 0.8\%$.

13. The battery of claim 10, wherein the positive electrode comprises the current collector layer and the active layer, the current collector layer of the positive electrode is selected from aluminum foil, and the active particles of the positive electrode are selected from one or more of lithium cobalt oxide, lithium nickel oxide, lithium manganese oxide, lithium nickel manganese oxide, lithium nickel cobalt manganese oxide, and lithium nickel cobalt aluminum oxide.

14. The battery of claim 10, wherein the negative electrode comprises the current collector layer and the active layer, the current collector layer of the negative electrode is selected from copper foil, and the active particles of the negative electrode are selected from one or more of graphite, meso-carbon micro-bead, hard carbon, and soft carbon.

15. The battery of claim 10, wherein each of the capillary pores has a radial dimension d satisfying: $2\text{ nm} \leq d \leq 30\text{ nm}$.

16. The battery of claim 10, wherein in the shell, the capillary pores have a distribution density a satisfying: $0.4\text{ cm}^3/\text{g} \leq a \leq 3\text{ cm}^3/\text{g}$.

17. The battery of claim 10, wherein each of the plurality of sub-particles has a particle size $D1$ satisfying: $50\text{ nm} \leq D1 \leq 90\text{ nm}$.

18. A power consuming device, comprising:

a device body; and

a battery, wherein the battery is configured to supply power to the device body, and the battery comprises:

a positive electrode;

a separator, disposed at one side of the positive electrode;

a negative electrode, disposed at one side of the separator positioned facing away from the positive electrode, wherein at least one of the positive electrode or the negative electrode is an electrode, and the electrode comprises:

a current collector layer; and

an active layer, disposed on a surface of the current collector layer, wherein the active layer comprises active particles and electrolyte retention particles; and each of the electrolyte retention particles comprises a core and a shell, the core is coated with the shell, the shell comprises a plurality of sub-particles, and each of the plurality of sub-particles has capillary pores; and

an electrolyte, wherein at least a part of the positive electrode and at least a part of the negative electrode are immersed in the electrolyte.

19. The power consuming device of claim 18, wherein the core has a diameter $D2$, and each of the electrolyte retention particles has an outer diameter $D3$, $D2$ and $D3$ satisfying: $1:3 \leq D2:D3 \leq 1:1.2$.

$D3 \leq 1:1.2$.

20. The power consuming device of claim **18**, wherein the core is a polymer synthesized from one or more selected from a group consisting of acrylic, acrylate, acrylamide, acrylonitrile, butadiene, styrene, and vinylidene fluoride.

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